Big Data Analytics Assignment 1 Report

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GitHub Repository: https://github.com/alexandercantrell/BigData2026-Assignment1

Question 1:

a. Report the accuracy of the **original** Llama-3.2-1B on both GSM8K and MATH datasets.
See Table 1

b. Fine-tune Llama-3.2-1B by setting the target rank to **16, 64, and 256**, respectively, and report the accuracy on both GSM8K and MATH datasets.
See Table 1

Accuracy	Original LLM	Fine-Tuned LLM (RANK=16)	Fine-Tuned LLM (RANK=64)	Fine-Tuned LLM (RANK=256)
GSM8K	3.5%	15.6%	20.5%	27.3%
MATH	0.1%	3.5%	4.7%	5.5%

Table 1

c. Provide your **observations and explanations** on the performance trends with different target ranks for both datasets.
From what I observed, fine-tuning increases accuracy significantly across both datasets with GSM8K going up from 3.5% to 15.6-27.3% and MATH from 0.1% to 3.5-5.5%. Higher ranks gave higher accuracy on both sets, likely because with a higher rank you get more trainable parameters, so the model has more capacity to adapt. The MATH dataset remained lower than GSM8K likely because the problems are harder with more complex answer formats.

Question 2:

a. Following Question 1, run experiments with target ranks of 16, 64, and 256, and record the **training time (minute)** for each setting.
See Table 2

	Fine-Tuned LLM (RANK=16)	Fine-Tuned LLM (RANK=64)	Fine-Tuned LLM (RANK=256)
Training Time (min.)	64.72	69.34	94.12

Table 2

b. Provide your **observations and explanations** on the training-time trends with different target ranks.
My main observation is that training time rises with rank (going from rank 16 to 64 or 64 to 256 jumps by 4.6 minutes and 24.8 minutes respectively). It isn't proportional though because most of the time is spent on the forward/backward pass through the frozen base model which is identical for every rank.

Question 3:

a. For each dataset, perform SVD and fine-tune the model by selecting the **four ranges of singular values** while fixing **Target Rank to 16**, and report the corresponding **accuracy in bar charts.**
(1.) The **Top** 16 singular values (set "component" to "default", which already exists in codes)
(2.) 16 consecutive singular values starting at the **25%** position

(3.) 16 consecutive singular values starting at the **50%** position
(4.) The **Bottom** 16 singular values
See Figure 1

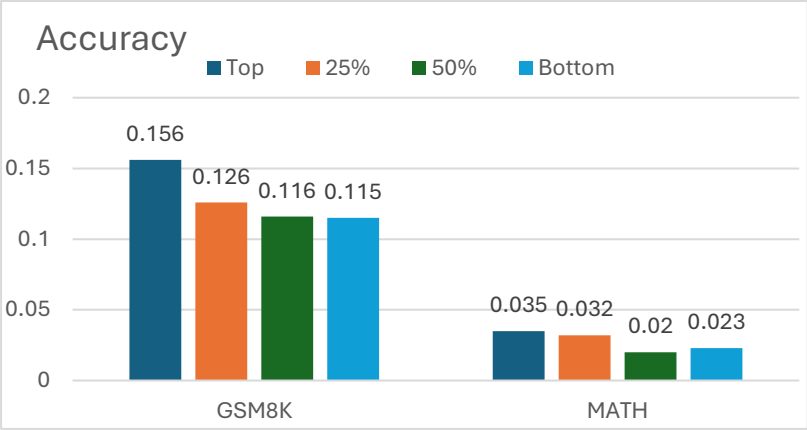


Figure 1

b. Provide your **observations and explanations** on the performance trends for both datasets
The main thing I observe is that top 16 is best on both datasets, and accuracy generally drops as you go lower into the set of singular values you choose (with an exception for MATH at the bottom). This is likely because the top singular values carry the largest share of the energy of the weight matrix as seen in Q4, so using those singular values allows for faster and better updates to the weights. However, even the bottom 16 singular values still beat out the model without training.

Question 4:

No fine-tuning or evaluation is required. Analyze only the original Llama-3.2-1B model.

a. Perform SVD on the **query projection ("q_proj") weights** of the LLM layers **0, 7, 15**:
    a. No need to specify target rank

See Table 3 and Figure 2

b. For each layer, please report the **minimum target rank** required to retain **at least 50% cumulative energy.**
See Table 3

	Layer 0	Layer 7	Layer 15
Minimum Target Rank for query projection weights	62	120	84

Table 3

c. Please provide the following **visualization for Layer 0, 7, 15, respectively:**
(For clarity, visualize only the **first 200** singular values):
    a. Cumulative energy (%) using a **line chart.**
    b. Singular value magnitudes using a **bar chart.**

See Figure 2

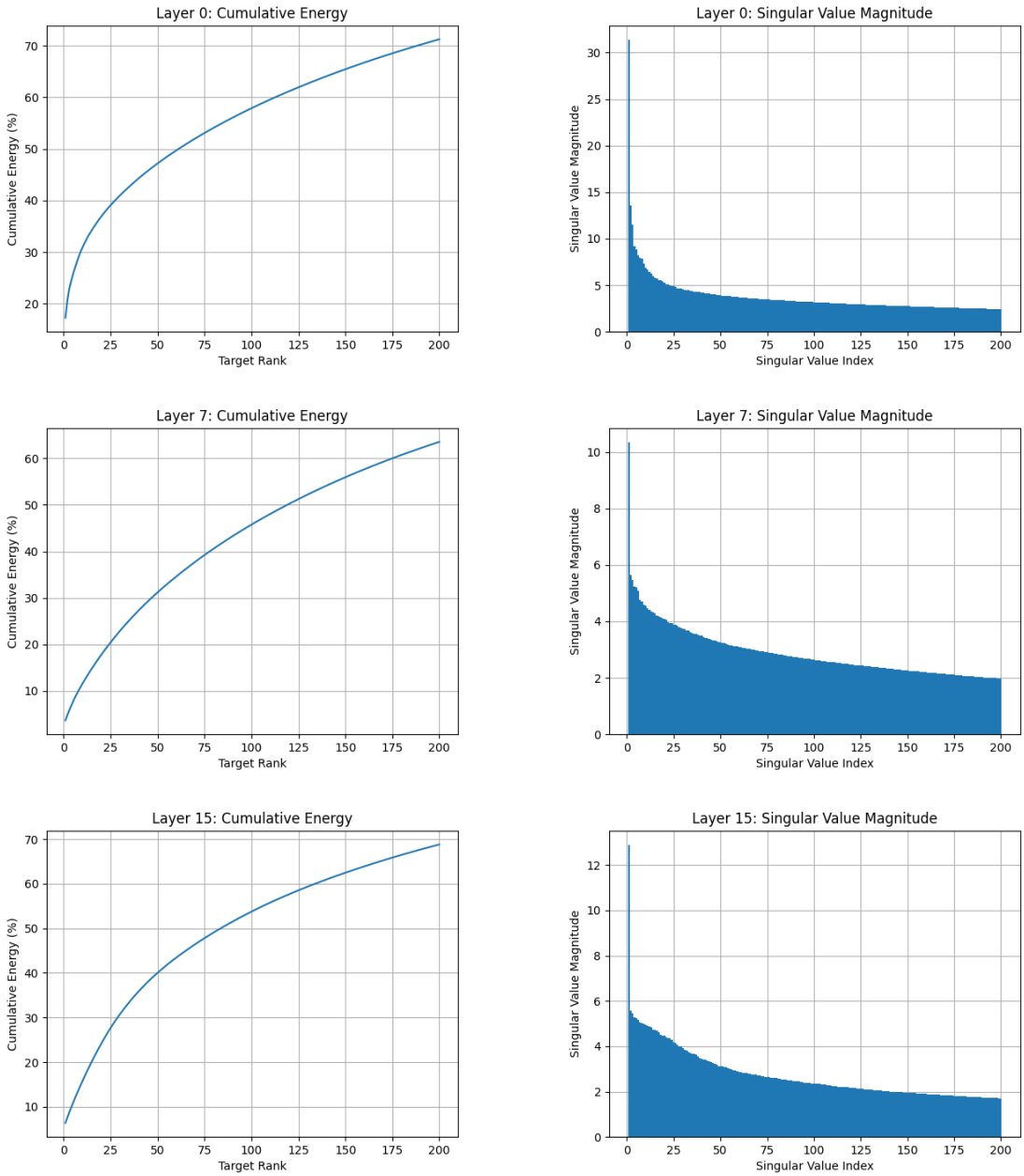


Figure 2

d. Provide your **observations** for the above results

The minimum ranks to retain 50% for layers 0, 7, and 15 are 62, 120, and 84 respectively all of which are <6% of the full rank (2048). This shows that over half the energy sits in a small number of singular values. However, in each graph, the first 200 singular values hold at most around 70% of the energy showing the matrices aren't that low rank. As for the individual layers, layer 0 appears to be most concentrated from both the cumulative energy graph and the fact that its top singular value holds a higher percent of the energy. Layer 7 is the most spread out as can be seen by its graph ending at the lowest cumulative energy and the 50% cutoff being so much higher. Finally, layer 15 is right in between both. It is notable though that the first and largest singular value of each matrix is significantly higher than all others.